

$$\int \frac{3x + 5 \sin e}{e^x} dx = 5 \int e^{-x} dx$$

$$= \left[1) \int \frac{x}{e^x} dx = \int \frac{u=x \Rightarrow u'=1}{v'=e^x \Rightarrow v=-\frac{1}{e^x}} \right] = x \left(-\frac{1}{e^x} \right) -$$

$$- \int -\frac{1}{e^x} dx = -\frac{x}{e^x} + \int \frac{1}{e^x} dx = -\frac{x}{e^x} - \frac{1}{e^x} + C =$$

$$= -\frac{x+1}{e^x} + C$$

$$2) \int \frac{\sin \frac{1}{e^x}}{e^x} dx = \left[t = \frac{1}{e^x} \Rightarrow dt = d\left(\frac{1}{e^x}\right) = -\frac{e^{-x}}{(e^x)^2} dx = -\frac{1}{e^x} dx \right]$$

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Упражнение 3

8.3.19

$$\int \frac{5 dx}{x + \sqrt{2}} = 5 \int \frac{dx}{x + \sqrt{2}} = \left[d(x + \sqrt{2}) = dx \right] = 5 \int \frac{d(x + \sqrt{2})}{x + \sqrt{2}} =$$

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Интервал I нуля, так 1) $x - \frac{1}{2}$ 1 степень

2) $x - a - \frac{1}{2}$ 1 степень

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Задача 3

8.3.19

$$\int \frac{5dx}{x+\sqrt{2}} = 5 \int \frac{dx}{x+\sqrt{2}} = \left[d(x+\sqrt{2}) = dx \right] = 5 \int \frac{d(x+\sqrt{2})}{x+\sqrt{2}} =$$

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2) $x - a - \frac{1}{2}$ 1 степень

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